

Exercise 7: Implementing the Observer Pattern

CODE:

```
import java.util.ArrayList;

interface Observer {

    void update(double price);

}

interface Stock {

    void registerObserver(Observer observer);

    void removeObserver(Observer observer);

    void notifyObservers();

}

// Concrete Subject

class StockMarket implements Stock {

    private ArrayList<Observer> observers = new ArrayList<>();

    private double stockPrice;

    public void registerObserver(Observer observer) {

        observers.add(observer);

    }

    public void removeObserver(Observer observer) {

        observers.remove(observer);

    }

    public void setStockPrice(double price) {

        stockPrice = price;

        notifyObservers();

    }

    public void notifyObservers() {

        for (Observer observer : observers) {

            observer.update(stockPrice);

        }

    }

}
```

```

    }
}

// Concrete Observer - Mobile App
class MobileApp implements Observer {
    public void update(double price) {
        System.out.println("Mobile App: Stock Price Updated to Rs." + price);
    }
}

// Concrete Observer - Web App
class WebApp implements Observer {
    public void update(double price) {
        System.out.println("Web App: Stock Price Updated to Rs." + price);
    }
}

// Main Class
public class Main {
    public static void main(String[] args) {
        StockMarket stockMarket = new StockMarket();
        Observer mobile = new MobileApp();
        Observer web = new WebApp();
        stockMarket.registerObserver(mobile);
        stockMarket.registerObserver(web);
        System.out.println("Stock Price Changed to Rs.2500");
        stockMarket.setStockPrice(2500);
        System.out.println();
        System.out.println("Stock Price Changed to Rs.2750");
        stockMarket.setStockPrice(2750);
    }
}

```

OUTPUT:

```
Stock Price Changed to Rs.2500
Mobile App: Stock Price Updated to Rs.2500.0
Web App: Stock Price Updated to Rs.2500.0

Stock Price Changed to Rs.2750
Mobile App: Stock Price Updated to Rs.2750.0
Web App: Stock Price Updated to Rs.2750.0
```